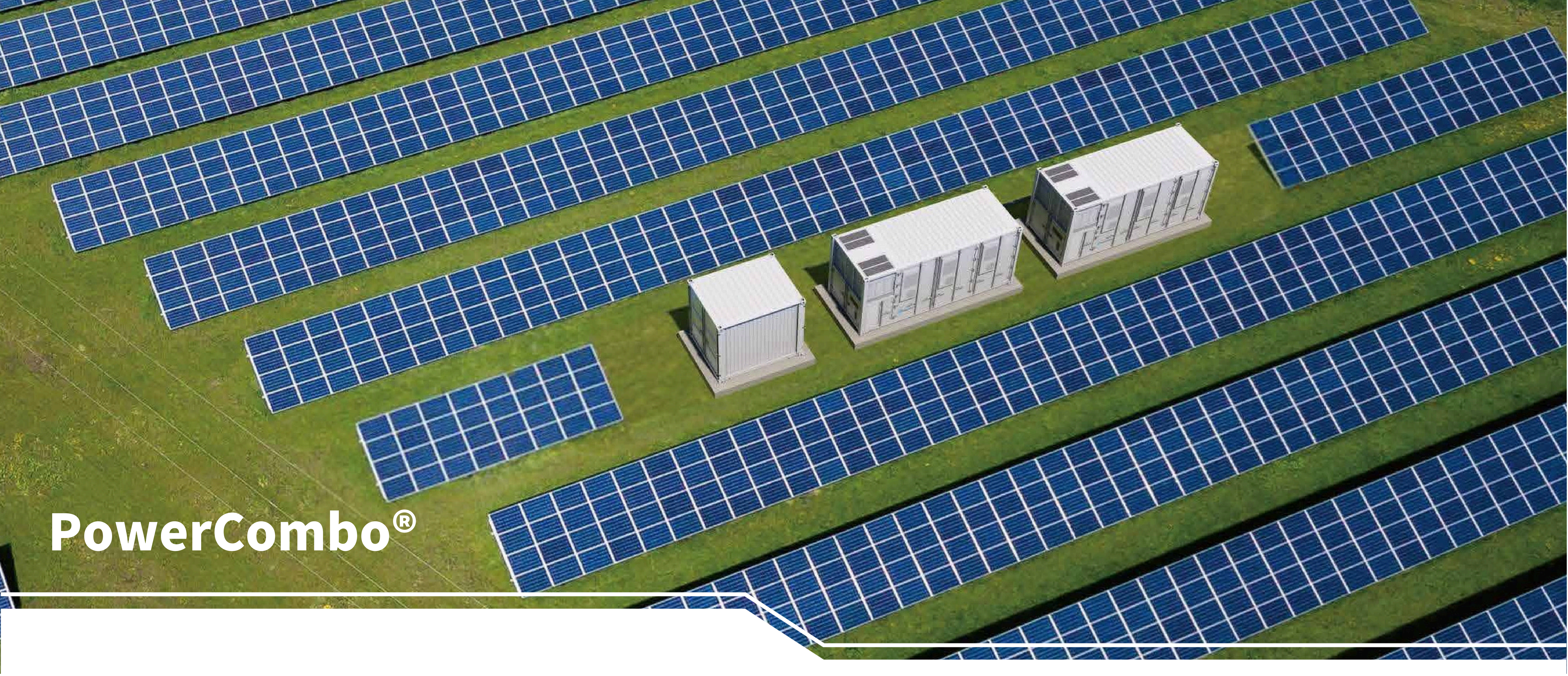
PowerCombo Datasheet

Available Reliable Accountable

www.unicom-bess-systems.com

info@unicom-bess-systems.com



PowerCombo®

PowerCombo, a high-performance, all-in-one, containerized battery energy storage system developed by Unicom, provides large-scale applications with intelligent and reliable BESS for auxiliary services etc. As a leading BESS product, PowerCombo is certificated by UL9540, UL1973, UL9540A, IEC62619, IEC 62933, IEC 62477, CE MARK, UN3536, NFPA855, bankability report, provides secure, reliable and safe power supply.

- **Auxiliary Service**
- **Behind-The-Meter**
- **Front-The-Meter**
- **Maximum Demand Management**
- **Peak Shaving**
- **EV Charging Power Supplyment**

CE Mark by TUV | **Bankability Report by DNV**

UL9540 UL9540A | **UN38.3 UN3536** | **NFPA855**

IEC62933	IEC62619	IEC62477	EN50549
IEEE1547	UL1973	UL1741	G99

Smart and User-friendly

Frequency / Voltage Stability; Grid Following / Grid Forming; Cloud dispatching and operation data analysis;

Economical and High Availability

Supports peak shaving, ADR, FCR/aFRR applications and more; Supports remote software update and components upgrade, lower operation and maintenance costs;

Safe and Reliable

Fault processing mechanism, responds to multiple fault scenarios; Customized BMS provides comprehensive measurement and protection functions;

Flexible and Convenient

Modular design, single unit management; Plug-and-play, supporting zero-test on site.

PowerCombo-20C2H1600K

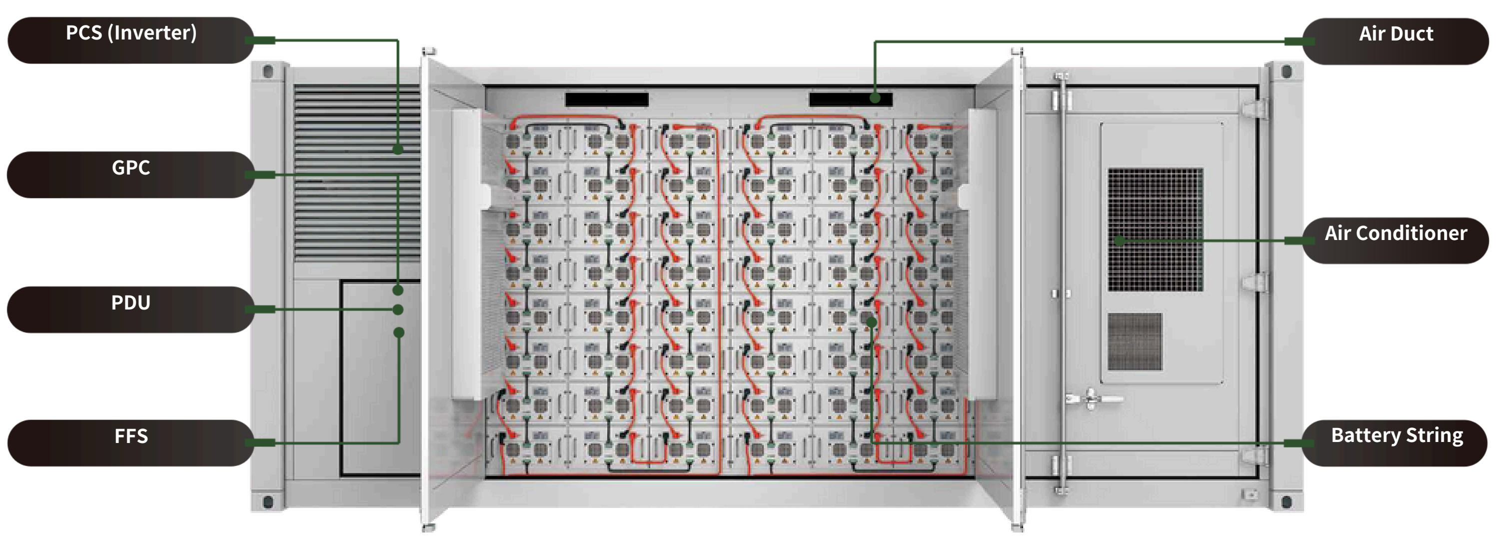
20C2H1600K-8LS385-8LP48

- Container Size
- Minimum Discharge Hour
- AC Power
- Quantity of Battery String
- Code of Battery String
- Quantity of Pack per String
- 48kWh/Pack

System Technical Specifications

Item	20C2H1600K	20C2H1000KLS
DC Data		
Battery chemistry	Lithium Iron Phosphate (LFP)	
Cell life cycle	8,000 cycles with 70% retention @0.5C 25℃	
Cell spec	3.2V/314Ah	
String configuration	1P384S	1P288S
Number of strings	8	8
DC rated energy capacity	3,086kWh	2,315kWh
Rated voltage	1,228.8V	921.6V
Voltage range	1,075.2V~1,363.2V	806.4V~1,022.4V
BMS communication interface	RS485, Ethernet	
BMS communication protocol	Modbus RTU, Modbus TCP	
AC Data	Standard Output	
Rated AC Power	1,600kW	1,000kW
Maximum AC power	1,760kW	1,100kW
Rated voltage	690V	400V
Grid voltage range	586.5~759V(Optional)	340~440V(Optional)
AC rate of current	1,338.8A	1,443.4A
Output THDi	<3%	
AC PF	-1~+1	
AC output	3W+PE	
General Data		
Dimension w/o clearances (L*W*H)	6,058x2,438x2,896mm	
Weight of the whole system	≤31t	≤26.1t
Degree of protection	IP54	
Operating temperature range	-30~50℃	
Relative humidity	0~95% (non-condensing)	
Max working altitude	2,000m/6,562feet(non-derating)	
Cooling concept of DC hatch	Liquid Cooling and Dehumidification system	
Fire fighting system	FK-5-1-12/Water fire fighting	
Communication interfaces	RS485, Ethernet	
Certificates	UL9540, IEC62933, UN3536, CE MARK by TÜV Rheinland	

PowerCombo-20C2H1200K



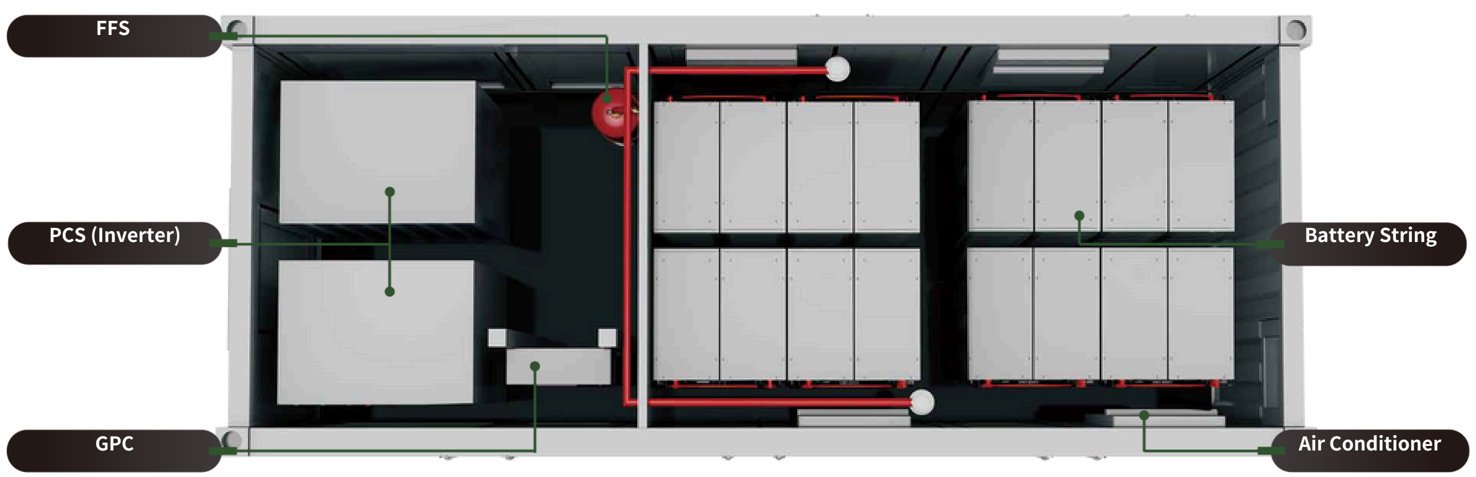
20C2H1200K-6S418-24P16

Container Size | Minimum Discharge Hours | AC Power | Quantity of Battery Strings | Code of Battery Strings | Quantity of Pack per String | 16kWh/Pack

System Technical Specifications

Item	20C2H1200K
DC Data	
Battery chemistry	Lithium Iron Phosphate (LFP)
Cell life cycle	8,000 cycles with 70% retention @ 0.5C 25°C
Cell spec	3.2V/314Ah
String configuration	1P384S
Number of strings	6
DC rated energy capacity	2,315kWh
Rated voltage	1,228.8V
Voltage range	1,075.2V~1,363.2V
BMS communication interface	RS485, Ethernet
BMS communication protocol	Modbus RTU, Modbus TCP
AC Data	
Rated AC Power	
Maximum AC power	1,200kW
Rated voltage	1,320kW
Grid voltage range	690V
AC rate of current	586.5~759V(configurable)
Output THDi	1,004.1A
AC PF	<3%
AC output	-1~+1
General Data	3W+PE
Dimension w/o clearances (W*D*H)	
Weight of the whole system	6,058x2,438x2,591mm
Degree of protection	26.5t
Operating temperature range	IP54
Relative humidity	-35~40°C
Max working altitude	0~95% (non-condensing)
Cooling concept of DC hatch	2,000m/6,562feet (non-derating)
Fire fighting system	Air cooling
Communication interfaces	FK-5-1-12/**Water fire fighting**
Certificates	RS485, Ethernet
	UL9540, IEC 62933, UN3536, CE Mark by TÜV

PowerCombo-20C1H1000K



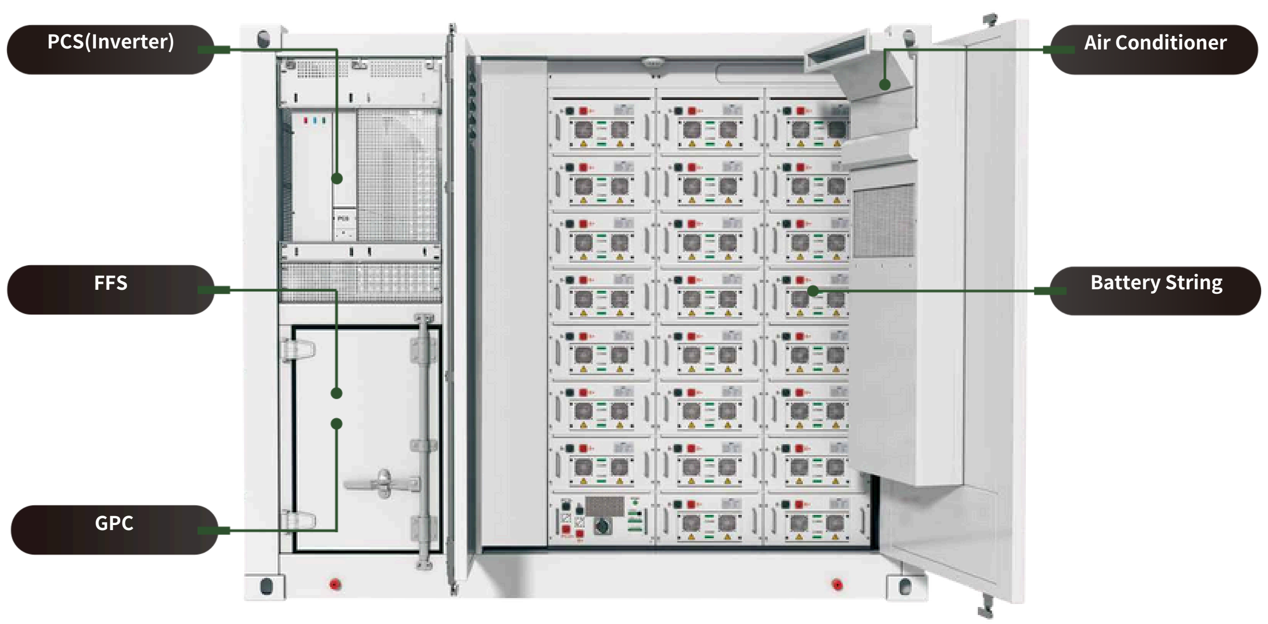
20C1H1000K-8S138-15P9

- Container Size
- Minimum Discharge Hours
- AC Power
- Quantity of Battery Strings
- Code of Battery Strings
- Quantity of Pack per String
- 9kWh/Pack

System Technical Specifications

Item	20C1H1000K
DC Data	
Battery chemistry	Lithium Iron Phosphate (LFP)
Cell life cycle	**6,000 cycles with 80% Retention @1C 25°C**
Cell spec	3.2V/90Ah
String configuration	2P240S
Number of strings	8
DC rated energy capacity	1,105kWh
Rated voltage	768V
Voltage range	672V~852V
BMS communication interface	RS485, Ethernet
BMS communication protocol	Modbus RTU, Modbus TCP
AC Data	
Rated AC power	1000kW 1100kW 400V
Maximum AC power	360~440V
Rated voltage	**(configurable)**
Grid voltage range	**1443.4A** <3% -1~+1
AC rate of current	3W+PE
Output THDi	
AC PF	
AC output	
General Data	
Dimension w/o clearances (W*D*H)	6,058x2,438x2,591mm
Weight of the whole system	19.5t
Degree of protection	IP54
Operating temperature range	-30~55°C
Relative humidity	0~95% (non-condensing)
Max working altitude	2,000m/6,562feet (non-derating)
Cooling concept of DC hatch	Air cooling
Fire fighting system	FK-5-1-12/**Water fire fighting**
Communication interfaces	RS485, Ethernet
Certificates	UL9540, UN3536

PowerCombo-10C2H360K



10C2H360K-2S418-23P16

Container Size | Minimum Discharge Hour | AC Power | Quantity of Battery String | Code of Battery String | Quantity of Pack per String | 16kWh/Pack

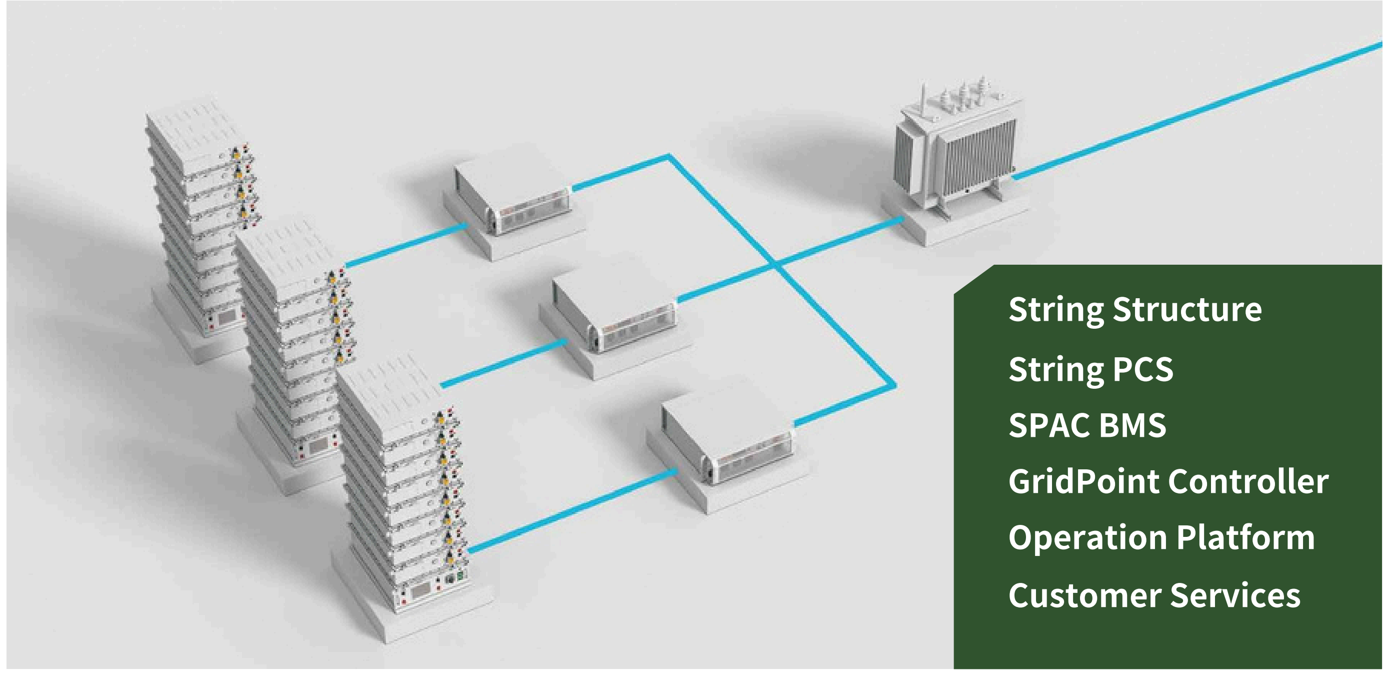
System Technical Specifications

Item	10C2H250K	10C2H360K
DC Data Battery chemistry		
Cell life cycle Cell spec String	Lithium Iron Phosphate (LFP)	
configuration	8,000 cycles with 70% retention @0.5C 25℃	
Number of strings	3.2V/314Ah	
Rated energy capacity	1P320S	1P368S
Rated voltage Voltage range	2	
BMS communication	643kWh	740kWh
interface BMS	1,024V	1,177.6V
communication protocol	896V~1,136V	1,030.4V~1,306.4V
AC Data	RS485, Ethernet	
Rated AC power	Modbus RTU, Modbus TCP	
Maximum AC power		
Rated voltage	250kW	360k
Grid voltage range	275kW	W
AC rate of current Output	400V	396k
THDi AC PF	-15%~+10% (configurable)	W
AC output	360.9A	600V 501.2A
General Data	<3%	
Dimension(L*W*H)	-1~+1	
Weight of the whole system	3W+PE	
Degree of protection		
Operating temperature	2,991x2,438x2,591mm	
range		
Relative humidity	9.2t	9.9t
Max working altitude	IP54	
Cooling concept of DC hatch	-30~55℃	
Fire fighting system	0~95% (non-condensing)	
Communication interfaces	2,000m/6,562feet (non-derating)	
Certificates	Air cooling	
	FK-5-1-12/Water fire fighting	
	RS485, Ethernet	
	UL9540, UN3536	

Advanced Technologies

Unicom not only leads in system integration but also has extraordinary capacities in innovation and developing leading BESS technologies.

By virtue of self-developed String Structure BESS, SPAC BMS and O&M System, Unicom provides products with high energy density, intelligence, availability and low fault rate. Maximizing the project revenue and guaranteeing the return on investment under the premise of safe and stable operation of BESS projects, acquiring more value and revenue from the BESS deployment.



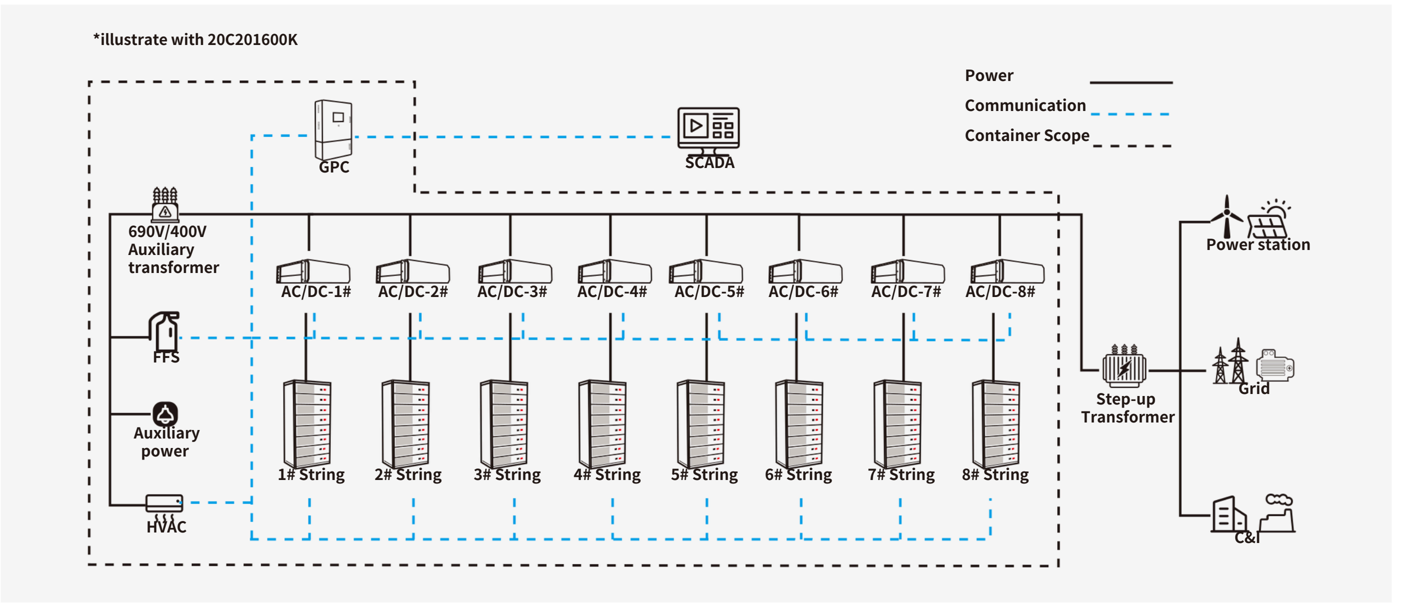
String Structure

String Structure BESS is a system design that leads each battery string to connect to its own modular PCS individually. Under this topology, each string is an independent unit-structure in DC circuitry. Charging and discharging of each battery string are independent, leads to smaller short-circuit current, and branch faults do not affect the operation of the entire BESS.

Features

- Guaranteeing High Availability
- Long System Lifespan
- Simple O&M

Topology



Advantages

- One-on-one Design
- Extreme Safety
- Modular Design
- Intelligent Battery Management
- Distributed Thermal Control
- Grid Forming

String PCS

Features

- Three-level topology, up to 99% conversion efficiency, better power quality.
- Supports constant power, constant current, constant voltage control, FCR, VSG, black start and other functions.
- Integrated string-level battery management function, high battery availability.
- Support IEC61850 protocol and have ms-level response capability.
- Intelligent multi-stage fan speed regulation, wide temperature operation capability, no derating at 50°C.
- IP66 protection level, can be installed vertically or horizontally outdoors.

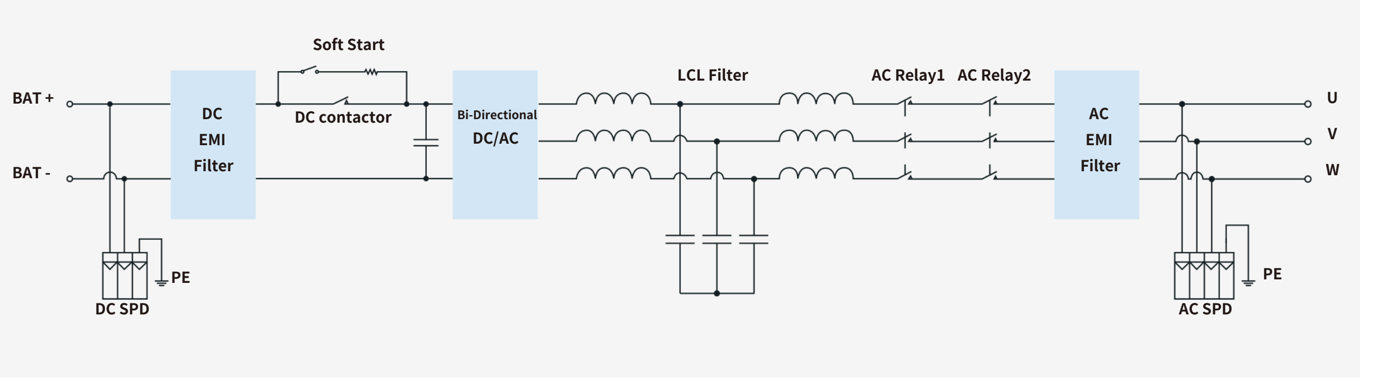
System Technical Specifications

Item	CBAC-125-HA-M	CBAC-140-HA-M-US	EH-0200-HA-M
DC Data Maximum			
DC voltage Maximum	1,500V	1,500V	1,500V
DC power Maximum	140kW	157kW	224kW 224A
DC current DC	234A	224A	1,000~1,400
voltage range AC	600~1,100	700~1,200	V
Data Rated AC power	V	V	
Maximum AC power	125kW	140k	200k
Maximum AC current	137kW	W	W
Rated grid voltage	198.4A	154k	220k
Grid voltage range	400V	W	W
Rated grid frequency	3W+PE ,	185.2-15%~+10%	184.1
PF adjustable range	(Configurable) 50/60Hz A -1~+1 <		A
THD (rated power)	3%	480V	690A

- Single-stage three-level modularization;
- Multi-branch input to reduce battery series and parallels connection;
- One-to-one Management.

System Features	
Dimensions (W*H*D)	810×845×275mm
weight	98kg
Isolation method	Non-isolated type (external transformer)
Maximum efficiency	99%
Protection level	IP66
Operating temperature range	-40~+60°C
Allowable humidity range	0~100%(No condensation)
Maximum operating altitude	4000m(>3000m Derating)
Cooling method	Temperature controlled forced air cooling
Communication interface	RS485/CAN/Ethernet

Circuit Diagram

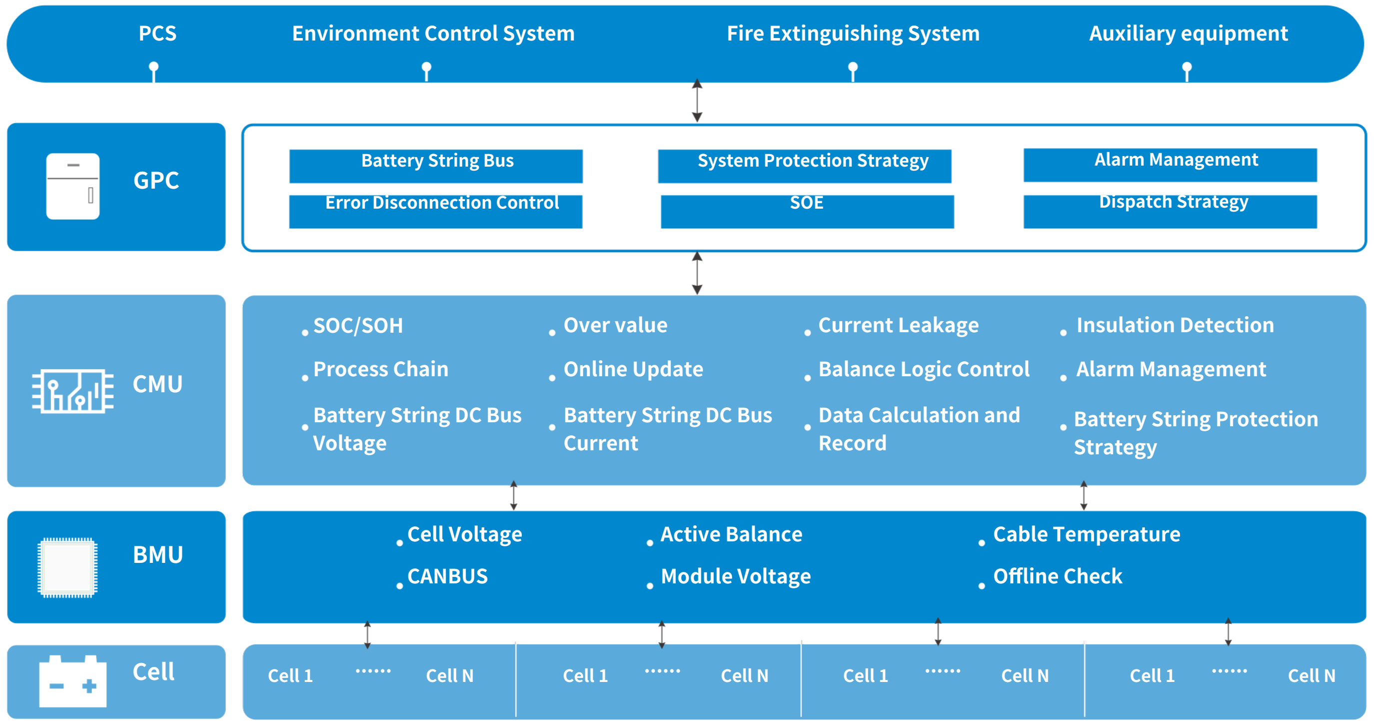


SPAC BMS

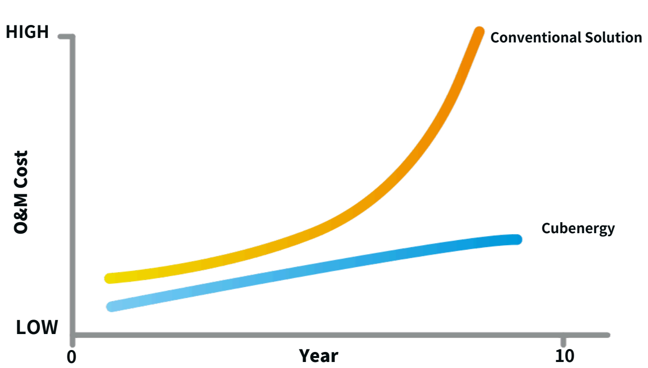
SPAC BMS technology is developed by Unicom, provides strong battery management ability to actively eliminate the imbalanced between battery cells in operation, further guarantee- ing the overall availability of BESS in multiple circles while ensuring high safety.

Functions

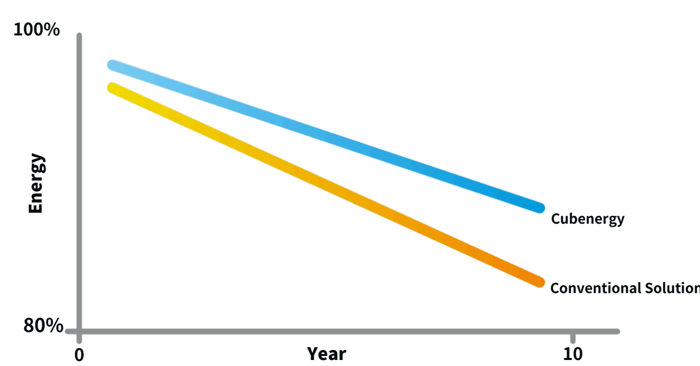
1. Monitor and protect safety of battery cell
2. Monitor and protect safety of electric system
3. Calculate and manage charging and discharging status
4. Calibrate and manage available energy
5. Optimize balance and manage cell consistency



Advantages



Maintenance Cost



Performance

GridPoint Controller

GridPoint Controlle（rGPC)

an advanced one-stop BESS control and status information and management center.

Power interface	Communication
AC400V/DC24V	Modubus TCP

Super Easy for BESS Power Plant Controller Integrating

Multi-function to shorten the FAT and SAT with any EMS and SCADA

Best tools for on-site opera- tion and maintenance

System Diagram